Longevity orthologs across four major model organisms

Complete visual catalogue • 320 genes • 206 direct pairs

Reviewed example = a checked experiment; GenAge annotation = catalogue evidence only.

All genes in the qualifying pair set are included; rows organise direct links, not proven conserved mechanisms.

	Budding yeast	Nematode	Fruit fly	Mouse
C054 **mev-1** 2 genes / 2 species 1 direct pair	No gene in this candidate network	**W1 mev-1** GenAge 435 **GenAge annotation** **Pro-Longevity** Lifespan: Decrease Refs: 9716135, 15269213	**F1 SdhC** GenAge 594 **GenAge annotation** **Pro-Longevity** Lifespan: Decrease Refs: 17854771	No gene in this candidate network
Direct pairs: W1–F1 (1:1)				
C117 **MRS4** 2 genes / 2 species 1 direct pair	**Y1 MRS4** GenAge 1439 **GenAge annotation** **Necessary for fitness** Lifespan: Decrease Refs: 18340043, 19030232	**W1 mfn-1** GenAge 1750 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 22253756	No gene in this candidate network	No gene in this candidate network
Direct pairs: W1–Y1 (1:m)				
C055 **MSH2** 2 genes / 2 species 1 direct pair	**Y1 MSH2** GenAge 1371 **GenAge annotation** **Necessary for fitness** Lifespan: Decrease Refs: 20883526	No gene in this candidate network	No gene in this candidate network	**M1 Msh2** GenAge 444 **GenAge annotation** **Pro-Longevity** Lifespan: Decrease Refs: 12629043
Direct pairs: M1–Y1 (1:1)				
C056 **MXR1** 2 genes / 2 species 1 direct pair	**Y1 MXR1** GenAge 450 **GenAge annotation** **Pro-Longevity** Lifespan: Increase + Decrease Refs: 15141092, 19049972	No gene in this candidate network	No gene in this candidate network	**M1 Msra** GenAge 446 **GenAge annotation** **Pro-Longevity** Lifespan: Decrease Refs: 11606777, 19487311
Direct pairs: M1–Y1 (1:1)				
C050 **NAM2** 2 genes / 2 species 1 direct pair	**Y1 NAM2** GenAge 1259 **GenAge annotation** **Necessary for fitness** Lifespan: Decrease Refs: 20421943	**W1 lars-2** GenAge 417 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 12447374	No gene in this candidate network	No gene in this candidate network
Direct pairs: W1–Y1 (1:1)				

Y/W/F/M identify genes in yeast/worm/fly/mouse columns. Pair codes retain the OMA source orientation.
Refs are catalogue-linked PMIDs, not newly reviewed experiments. Empty cells do not establish biological absence.
GenAge lifespan labels do not specify interventions. Only reviewed examples have experiment arrows and colours.
Sources: GenAge Build 21; OMA All.May2026. See atlas legend, gene index and source tables for evidence details.